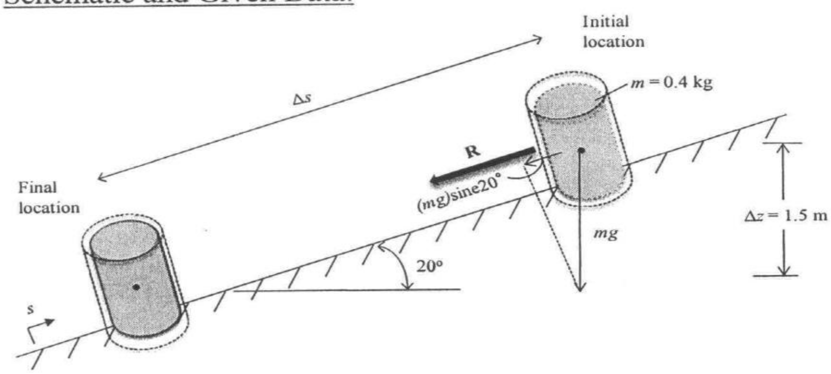
2.12 During the packaging process, a can of soda of mass 0.4 kg moves down a surface inclined at 20° relative to the horizontal, as shown in Fig. P2.12. The can is acted upon by a constant force **R** parallel to the incline and by the force of gravity. The magnitude of **R** is 0.05 N. Ignoring friction between the can and the inclined surface, determine the can's change in kinetic energy, in J, and whither it is *increasing* or *decreasing*. If friction between the can and the inclined surface were significant, what would effect would that have on the value of the change in kinetic energy? Let $g = 9.8\ \text{m/s}^2$.

Known: Can moves down a surface that is inclined relative to the horizontal. The can is acted upon by a constant force parallel to the incline and by the force of gravity.

Find: Can's change in kinetic energy, in J, and whether it is *increasing* or *decreasing*. If friction between the can and the inclined surface were significant, what effect would that have on the value of the change in kinetic energy?

Schematic and Given Data:



Engineering Model:
(1) The can is a closed system.
(2) The acceleration of gravity is constant.
(3) The applied force **R** is constant.
(4) Ignore friction between the can and inclined surface.

Analysis:

Begin with Eq. 2.6

$$\int_{s_1}^{s_2} \underline{F}\cdot d\underline{s} = \frac{1}{2}m\left(V_2^2 - V_1^2\right) = \Delta KE \qquad (1)$$

From the free body diagram in the schematic, the dot product can be expressed as

$$\underline{F}\cdot d\underline{s} = \left(\mathbf{R} + (mg)\text{sine}20^\circ\right)ds$$

Substituting into Eq. (1)

$$\int_{s_1}^{s_2} \underline{F}\cdot d\underline{s} = \int_{s_1}^{s_2} (\mathbf{R} + (m$$

Since Δz = Δs sine 20°, Eq. (2) becomes

$$\int_{s_1}^{s_2} (\mathbf{R})ds + (mg)\Delta z = (\mathbf{R})\Delta s + (mg)\Delta z = \Delta KE \qquad (3)$$

Evaluate Δs

$$\Delta s = \frac{\Delta z}{\text{sine}\,20^\circ} = \frac{1.5\ \text{m}}{0.342} = 4.39\ \text{m}$$

Problem 2.12 (Continued)

Subsituting all known and calculated data into Eq. (3)

$$\Delta KE = (0.05\text{N})(4.39\text{m})\left|\frac{1\text{J}}{1\text{N}\cdot\text{m}}\right| + (0.4\text{kg})\left(9.8\frac{\text{m}}{\text{s}^2}\right)(1.5\text{m})\left|\frac{1\text{N}}{1\frac{\text{kg}\cdot\text{m}}{\text{s}^2}}\right|\left|\frac{1\text{J}}{1\text{N}\cdot\text{m}}\right| =$$

#1

$$\Delta KE = 0.22\,\text{J} + 5.88\,\text{J} = 6.1\,\text{J}$$

Which corresponds to an *increase* in kinetic energy.

If friction were significant, the magnitude of the net force acting in the direction of motion would be less, and thus the kinetic energy change would be less than calculat__.

1. Observe that in the absence of the force **R** the can is acted on only by gravity, and the can's change in kinetic energy is 5.88 J.

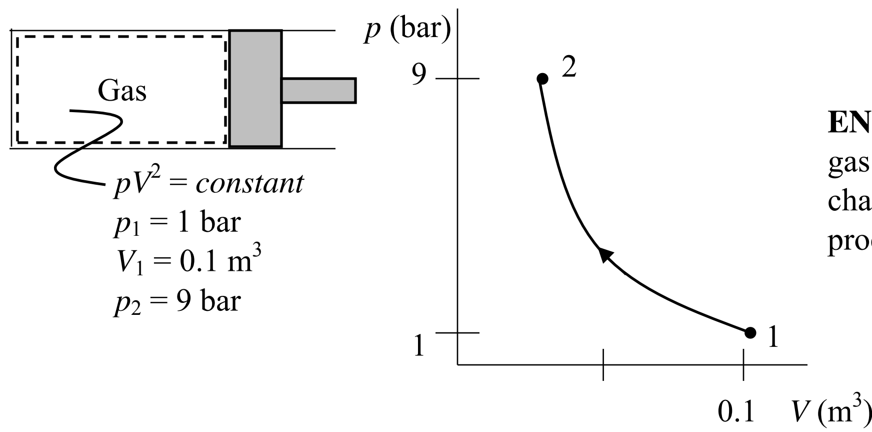
Problem 2.17

A gas in a piston-cylinder assembly undergoes a process for which the relationship between pressure and volume is $pV^2 = constant$. The initial pressure is 1 bar, the initial volume is 0.1 m^3, and the final pressure is 9 bar. Determine (a) the final volume, in m^3, and (b) the work for the process, in kJ.

KNOWN: A gas in a piston-cylinder assembly undergoes a process during which $pV^2 = constant$. State data are provided.

FIND: Determine the final volume occupied by the gas and the work for the process.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The gas is a closed system. (2) Volume change is the only work mode. (3) the process of the obeys $pV^2 = constant$.

ANALYSIS:

(a) We have $pV^2 = constant$. Thus $p_1V_1^2 = p_2V_2^2$. Solving for V_2

$$V_2 = \left[\frac{p_1}{p_2}\right]^{1/2} V_1 = \left[\frac{1}{9}\right]^{1/2} (0.1\ \text{m}^3) = 0.0333\ \text{m}^3 \leftarrow$$

(b) Using Eq. 2.17 to determine the work

$$W = \int_1^2 p\,dV = \frac{p_2V_2 - p_1V_1}{(1-2)} \qquad \text{(See Example 2.1(a) for the integration)}$$

Inserting values and converting units

$$W = \frac{(9\ \text{bar})(0.033\ \text{m}^3) - (1\ \text{bar})(0.1\ \text{m}^3)}{(-1)} \left|\frac{10^5\ \text{N/m}^2}{1\ \text{bar}}\right| \left|\frac{1\ \text{kJ}}{10^3\ \text{N}\cdot\text{m}}\right|$$

$$= -20\ \text{kJ} \leftarrow$$

Negative sign denotes energy transfer *to* the gas by work during the compression process.

Problem 2.25

Air contained within a piston-cylinder assembly undergoes three processes in series:

Process 1-2: Compression during which the pressure-volume relationship is $pV = constant$ from $p_1 = 10$ lbf/in.^{2}, $V_1 = 4$ ft^3 to $p_2 = 50$ lbf/in.^{2}

Process 2-3: Constant volume from state 2 to state 3 where $p = 10$ lbf/in.^{2}

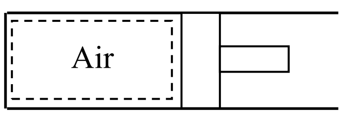
Process 3-1: Constant pressure expansion to the initial state.

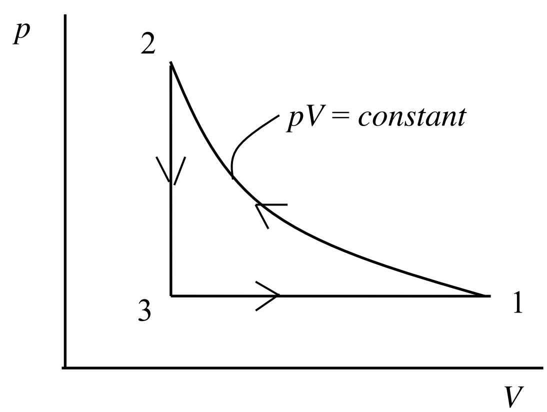
Sketch the processes in series on a p-V diagram. Evaluate (a) the volume at state 2, in ft^3, and (b) the work for each process, in Btu.

KNOWN: Air within a piston-cylinder assembly undergoes three processes in series.

FIND: Sketch the processes in series on a p-V diagram. Evaluate (a) the volume at state 2, and (b) the work for each process.

SCHEMATIC AND GIVEN DATA:





Process 1-2: Compression during which the pressure-volume relationship is $pV = constant$ from $p_1 = 10$ lbf/in.^{2}, $V_1 = 4$ ft^3 to $p_2 = 50$ lbf/in.^{2}
Process 2-3: Constant volume from state 2 to state 3 where $p = 10$ lbf/in.^{2}
Process 3-1: Constant pressure expansion to the initial state.

ENGINEERING MODEL: (1) The gas is the closed system. (2) Volume change is the only work mode. (3) Each of the three processes is specified.

ANALYSIS: (a) For process 1-2; $pV = constant$. Thus $p_1V_1 = p_2V_2$, and

$$V_2 = \left(\frac{p_1}{p_2}\right)V_1 = \left(\frac{10\text{lbf/in.}^2}{50\text{lbf/in.}^2}\right)(4\text{ ft}^3) = 0.8\text{ ft}^3 \longleftarrow$$

(b) Since volume change is the only work mode, Eq. 2.17 applies.

Process 1-2: For process 1-2, $pV = constant = p_1V_1$. Thus

$$W_{12} = \int_{V_1}^{V_2} pdV = \int_{V_1}^{V_2} \frac{C}{V}dV = C\ln\left(\frac{V_2}{V_1}\right) = (p_1V_1)\ln\left(\frac{V_2}{V_1}\right)$$

Problem 2.25 (Continued)

Inserting values and converting units

$$W_{12} = \left(10\ \frac{\text{lbf}}{\text{in.}^2}\right)(4\ \text{ft}^3)\ \ln\left(\frac{0.8\ \text{ft}^3}{4\ \text{ft}^3}\right)\left|\frac{144\ \text{in.}^2}{1\ \text{ft}^2}\right|\left|\frac{1\ \text{Btu}}{778\ \text{ft}\cdot\text{lbf}}\right| = -11.92\ \text{Btu (in)} \longleftarrow$$

Process 2-3: Constant volume (piston does not move). Thus $W_{23} = 0$ $\longleftarrow$

Process 3-1: Constant pressure processes ($p_3 = p_1$): $W_{31} = \int_{V_3}^{V_1} p dV = p_1(V_1 - V_3)$

Noting that $V_3 = V_2$

$$W_{31} = \left(10\ \frac{\text{lbf}}{\text{in}^2.}\right)(4 - 0.8)\text{ft}^3\left|\frac{144\ \text{in.}^2}{1\ \text{ft}^2}\right|\left|\frac{1\ \text{Btu}}{778\ \text{ft}\cdot\text{lbf}}\right| = 5.92\ \text{Btu (out)} \longleftarrow$$

---

1. The *net* work for the three process is

$W_{net} = W_{12} + W_{23} + W_{31}$ = (-11.92) + 0 + (5.92)) = - 6 kJ (*net* work is negative - in)

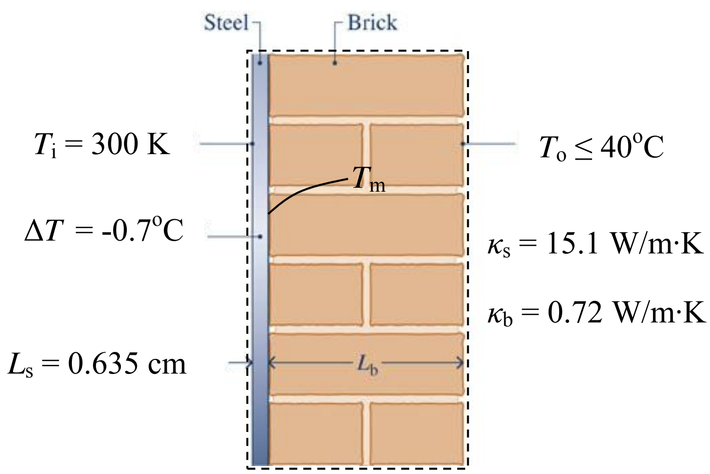
Problem 2.36

As shown in Fig. P2.36, an oven wall consists of a 0.635-cm-thick layer of steel (κ_s = 15.1 W/m·K) and a layer of brick (κ_b = 0.72 W/m·K). At steady state, a temperature decrease of 0.7°C occurs over the steel layer. The inner temperature of the steel layer is 300 K. If the temperature of the outer surface of the brick must be no greater than 40°C, determine the thickness of brick, in cm, that ensures this limit is met. What is the rate of conduction, in kW per m^2 of wall surface area?

KNOWN: Steady-state data are provided for a composite wall formed from a steel layer and a brick layer.

FIND: Determine the minimum thickness of the brick layer to keep the outer surface temperature of the brick at or below a specified value.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The wall is the system at steady state. (2) The temperature varies linearly through each layer.

ANALYSIS: Using Eq. 2.31 together with model assumption 2

$$\left(\frac{\dot{Q}}{A}\right)_{\text{steel}} = -\kappa_s\left[\frac{T_m - T_i}{L_s}\right] \quad \text{and} \quad \left(\frac{\dot{Q}}{A}\right)_{\text{brick}} = -\kappa_b\left[\frac{T_o - T_m}{L_b}\right]$$

where T_m denotes the temperature at the steel-brick interface.

At steady state, the rate of conduction *to* the interface through the steel must equal the rate of conduction *from* the interface through the brick: $(\dot{Q}/A)_{\text{steel}} = (\dot{Q}/A)_{\text{brick}}$. Thus

$$-\kappa_s\left[\frac{T_m - T_i}{L_s}\right] = -\kappa_b\left[\frac{T_o - T_m}{L_b}\right]$$

And solving for L_b we get

(300 – 0.7) = 299.3 °C

$$L_b = \frac{\kappa_b}{\kappa_s}\left[\frac{T_o - T_m}{\underline{T_m - T_i}}\right] L_s$$

= – 0.7 °C

<u>Problem 2.36 (Continued)</u>

$$L_b = \left(\frac{0.72\ \mathrm{W/m{\cdot}K}}{15.1\ \mathrm{W/m{\cdot}K}}\right)\left[\frac{299.3 - T_o}{0.7}\right](0.635\ \mathrm{cm})$$

Since $T_o \leq 40\ ^oC$

$$L_b \geq \left(\frac{0.72}{15.1}\right)\left[\frac{299.3 - 40}{0.7}\right](0.635\ \mathrm{cm})$$

$$L_b \geq 11.22\ \mathrm{cm} \longleftarrow$$

The rate of conduction is

$$\left(\frac{\dot{Q}}{A}\right)_{\mathrm{steel}} = -\kappa_s\left[\frac{T_m - T_i}{L_s}\right] = -(15.1\ \mathrm{W/m\cdot K})\left[\frac{299.3-300}{0.635\ \mathrm{cm}}\right]\left|\frac{100\ \mathrm{cm}}{1\ \mathrm{m}}\right|\left|\frac{1\ \mathrm{kW}}{10^3\mathrm{W}}\right| = 1.665\ \mathrm{kW/m^2} \longleftarrow$$

or

$$\left(\frac{\dot{Q}}{A}\right)_{\mathrm{brick}} = -\kappa_s\left[\frac{T_o - T_m}{L_b}\right] = -(0.72\ \mathrm{W/m\cdot K})\left[\frac{40-299.3}{11.22\ \mathrm{cm}}\right]\left|\frac{100\ \mathrm{cm}}{1\ \mathrm{m}}\right|\left|\frac{1\ \mathrm{kW}}{10^3\mathrm{W}}\right| = 1.664\ \mathrm{kW/m^2} \longleftarrow$$

The slight difference is due to round-off.

Problem 2.25

Air contained within a piston-cylinder assembly undergoes three processes in series:

Process 1-2: Compression during which the pressure-volume relationship is $pV = constant$ from p_1 = 10 lbf/in.^{2}, V_1 = 4 ft^3 to p_2 = 50 lbf/in.^{2}

Process 2-3: Constant volume from state 2 to state 3 where p = 10 lbf/in.^{2}

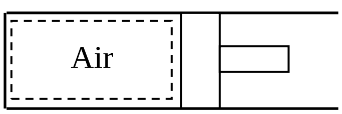
Process 3-1: Constant pressure expansion to the initial state.

Sketch the processes in series on a p-V diagram. Evaluate (a) the volume at state 2, in ft^3, and (b) the work for each process, in Btu.

KNOWN: Air within a piston-cylinder assembly undergoes three processes in series.

FIND: Sketch the processes in series on a p-V diagram. Evaluate (a) the volume at state 2, and (b) the work for each process.

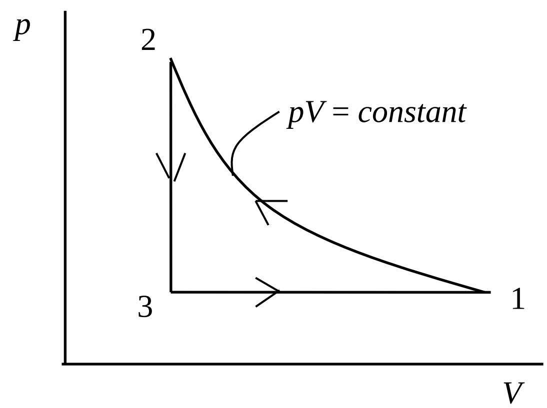
SCHEMATIC AND GIVEN DATA:



Process 1-2: Compression during which the pressure-volume relationship is $pV = constant$ from p_1 = 10 lbf/in.^{2}, V_1 = 4 ft^3 to p_2 = 50 lbf/in.^{2}

Process 2-3: Constant volume from state 2 to state 3 where p = 10 lbf/in.^{2}

Process 3-1: Constant pressure expansion to the initial state.



ENGINEERING MODEL: (1) The gas is the closed system. (2) Volume change is the only work mode. (3) Each of the three processes is specified.

ANALYSIS: (a) For process 1-2; $pV = constant$. Thus $p_1V_1 = p_2V_2$, and

$$V_2 = \left(\frac{p_1}{p_2}\right)V_1 = \left(\frac{10\,\text{lbf/in.}^2}{50\,\text{lbf/in.}^2}\right)(4\ \text{ft}^3) = 0.8\ \text{ft}^3$$

(b) Since volume change is the only work mode, Eq. 2.17 applies.

Process 1-2: For process 1-2, $pV = constant = p_1V_1$. Thus

$$W_{12} = \int_{V_1}^{V_2} p\,dV = \int_{V_1}^{V_2} \frac{C}{V}\,dV = C\ln\left(\frac{V_2}{V_1}\right) = (p_1V_1)\ln\left(\frac{V_2}{V_1}\right)$$

Problem 2.25 (Continued)

Inserting values and converting units

$$W_{12} = \left(10\ \frac{\text{lbf}}{\text{in.}^2}\right)(4\ \text{ft}^3)\ \ln\left(\frac{0.8\ \text{ft}^3}{4\ \text{ft}^3}\right)\left|\frac{144\ \text{in.}^2}{1\ \text{ft}^2}\right|\left|\frac{1\ \text{Btu}}{778\ \text{ft}\cdot\text{lbf}}\right| = -11.92\ \text{Btu (in)} \longleftarrow$$

Process 2-3: Constant volume (piston does not move). Thus $W_{23} = 0$ $\longleftarrow$

Process 3-1: Constant pressure processes ($p_3 = p_1$): $W_{31} = \int_{V_3}^{V_1} p dV = p_1(V_1 - V_3)$

Noting that $V_3 = V_2$

$$W_{31} = \left(10\,\frac{\text{lbf}}{\text{in}^2.}\right)(4 - 0.8)\text{ft}^3\left|\frac{144\ \text{in.}^2}{1\ \text{ft}^2}\right|\left|\frac{1\ \text{Btu}}{778\ \text{ft}\cdot\text{lbf}}\right| = 5.92\ \text{Btu (out)} \longleftarrow$$

---

1. The *net* work for the three process is

$W_{net} = W_{12} + W_{23} + W_{31}$ = (-11.92) + 0 + (5.92)) = - 6 kJ (*net* work is negative - in)

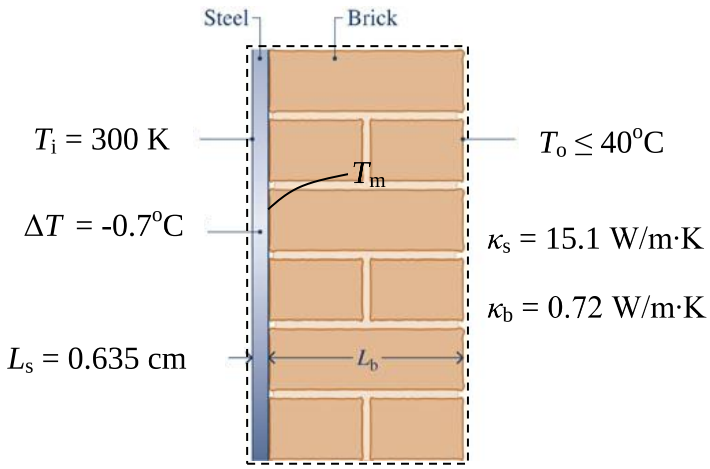
Problem 2.36

As shown in Fig. P2.36, an oven wall consists of a 0.635-cm-thick layer of steel (κ_s = 15.1 W/m·K) and a layer of brick (κ_b = 0.72 W/m·K). At steady state, a temperature decrease of 0.7°C occurs over the steel layer. The inner temperature of the steel layer is 300 K. If the temperature of the outer surface of the brick must be no greater than 40°C, determine the thickness of brick, in cm, that ensures this limit is met. What is the rate of conduction, in kW per m^2 of wall surface area?

KNOWN: Steady-state data are provided for a composite wall formed from a steel layer and a brick layer.

FIND: Determine the minimum thickness of the brick layer to keep the outer surface temperature of the brick at or below a specified value.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The wall is the system at steady state. (2) The temperature varies linearly through each layer.

ANALYSIS: Using Eq. 2.31 together with model assumption 2

$$\left(\frac{\dot{Q}}{A}\right)_{\text{steel}} = -\kappa_s\left[\frac{T_m - T_i}{L_s}\right] \quad \text{and} \quad \left(\frac{\dot{Q}}{A}\right)_{\text{brick}} = -\kappa_b\left[\frac{T_o - T_m}{L_b}\right]$$

where T_m denotes the temperature at the steel-brick interface.

At steady state, the rate of conduction *to* the interface through the steel must equal the rate of conduction *from* the interface through the brick: $(\dot{Q}/A)_{\text{steel}} = (\dot{Q}/A)_{\text{brick}}$. Thus

$$-\kappa_s\left[\frac{T_m - T_i}{L_s}\right] = -\kappa_b\left[\frac{T_o - T_m}{L_b}\right]$$

And solving for L_b we get

(300 – 0.7) = 299.3 °C

$$L_b = \frac{\kappa_b}{\kappa_s}\left[\frac{T_o - T_m}{T_m - T_i}\right] L_s$$

= – 0.7 °C

Problem 2.36 (Continued)

$$L_\mathrm{b} = \left(\frac{0.72\ \mathrm{W/m{\cdot}K}}{15.1\ \mathrm{W/m{\cdot}K}}\right)\left[\frac{299.3 - T_\mathrm{o}}{0.7}\right](0.635\ \mathrm{cm})$$

Since $T_\mathrm{o} \leq 40\ ^\mathrm{o}\mathrm{C}$

$$L_\mathrm{b} \geq \left(\frac{0.72}{15.1}\right)\left[\frac{299.3 - 40}{0.7}\right](0.635\ \mathrm{cm})$$

$$L_\mathrm{b} \geq 11.22\ \mathrm{cm} \longleftarrow$$

The rate of conduction is

$$\left(\frac{\dot{Q}}{\mathrm{A}}\right)_\mathrm{steel} = -\kappa_\mathrm{s}\left[\frac{T_\mathrm{m} - T_\mathrm{i}}{L_\mathrm{s}}\right] = -(15.1\ \mathrm{W/m \cdot K})\left[\frac{299.3 - 300}{0.635\ \mathrm{cm}}\right]\left|\frac{100\ \mathrm{cm}}{1\ \mathrm{m}}\right|\left|\frac{1\ \mathrm{kW}}{10^3\mathrm{W}}\right| = 1.665\ \mathrm{kW/m^2} \longleftarrow$$

or

$$\left(\frac{\dot{Q}}{\mathrm{A}}\right)_\mathrm{brick} = -\kappa_\mathrm{s}\left[\frac{T_\mathrm{o} - T_\mathrm{m}}{L_\mathrm{b}}\right] = -(0.72\ \mathrm{W/m \cdot K})\left[\frac{40 - 299.3}{11.22\ \mathrm{cm}}\right]\left|\frac{100\ \mathrm{cm}}{1\ \mathrm{m}}\right|\left|\frac{1\ \mathrm{kW}}{10^3\mathrm{W}}\right| = 1.664\ \mathrm{kW/m^2} \longleftarrow$$

The slight difference is due to round-off.